

Serotonin and the pursuit of motherhood: The impact of serotonin on the quality of
maternal care and anxiety in offspring.

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Abstract

Quality of maternal care impacts offspring development. The current study uses an animal-model of serotonin transporter (5-HTT) dysfunction, $SERT^{-/-}$ (knockout), $SERT^{+/-}$ (50% reduction) and $SERT^{+/+}$ (regular function), to explore a relationship between maternal care and 5-HTT. The quality of maternal care was measured through the mother's motivation to retrieve pups, mother and pup bond, and the adult offspring's anxiety-like behaviours.

$SERT^{-/-}$ mothers were less motivated to care for their pups, formed weak filial bonds, and their offspring showed high levels of anxiety-like behaviours. The $SERT^{+/-}$ and $SERT^{+/+}$ did not differ in their quality of maternal care, suggesting a relationship between the 5-HTT and maternal care. We speculate that the mechanism behind this relationship is due to a decrease in the rewarding elements of social interactions felt by the $SERT^{-/-}$. We suggest there is an optimal level of 5-HTT, as further increases have no effect on the quality of maternal care, shown by the similarity between the $SERT^{+/-}$ and $SERT^{+/+}$ genotypes.

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Emotion regulation is a complex process, which entails an intricate system of neural networks. Among these is the serotonergic system, and when it is disrupted has been implicated in the development of mood and anxiety disorders (Caspi et al., 2003). Early stressful life events influence how we interact with the world, the development of said disorders, and impacts how individuals learn to respond to stress. As a possible mechanism for early life stress the mother's maternal care of her offspring may have long term consequences on their development.

Maternal attachment and care

The forming of social bonds is an important aspect of social life which begins with the parent-child relationship. Filial or maternal attachment is described as the strong social bond formed between mother and offspring. This bond is important for offspring survival and reflects the level of care given to the offspring by the mother (Broad, Curley, & Keverne, 2006). It is seen in most animals whose young are born in a helpless state (Broad et al., 2006). Maternal attachment is formed through a number of mechanisms including neuronal, physiology (such as hormones like oxytocin), and social cues such as maternal care (Mogi, Nagasawa, & Kikusui, 2011). Attachment and level of maternal care scaffolds the development of emotional and social responses, such as reacting to stressful situations (Francis & Meaney, 1999).

Mammals tend to have fewer offspring than reptiles or fish, as such the offspring require a higher level of care to survive beyond infancy (Wu, Gao, Chou, Davis, & Li, 2016). A mother typically provides an infant's first social interactions and mediates their interactions with the outside world (Feldman, 2017). There is a relationship between a mother's psychopathologies and her children's developmental trajectory (Blandon,

Calkins, Keane, & O'Brien, 2008). Severe depressive symptoms in mothers can negatively impact their child's emotional regulation development. This is important for their ability to cope with stressful situations. Rodents require a considerable amount of maternal care (licking, nursing, nest building and retrieving stray pups) as their pups are born deaf, blind and unable to regulate their internal temperatures (Broad et al., 2006). This maternal care decreases as the pups develop, for instance nest building stops when the pups are able to regulate their body temperature (Maher, Barbash, Lynch, & Swoap, 2015).

The maternal bond has been suggested as a way of ensuring the social development of the infant through the quality of maternal care (Mogi et al., 2011). As different species require different skills and cognitive abilities in their social habitats, maternal care serves different functions based on developmental needs (Carter, 1998). A clear example of this is rats, who are born in burrows or nests. Pups have little experience with the outside world directly, their mothers provide a link to the outside world which informs her maternal care and their development (Francis & Meaney, 1999). For example, it is her response to stressors from the environment which informs how the pups learn to deal with stress, through the quality of maternal care. This is in contrast to other animals such as guinea pigs whom are precocial, and are comparatively less reliant on the mothers care (Hennessy, 2014).

Animal models of development show that maternal care affects everything from physiology to behaviour and cognition in offspring (Whishaw & Kolb, 2014). Pups from mothers who show a high level of care, through licking and grooming during the first week of life show reduced anxiety-and-depressive-like behaviours compared to those from mothers with low quality maternal care (Mogi et al., 2011). Maternal care has also been shown to effect neural development, such as the neuroendocrine system's response to stress (Caldji et al., 1998). When weaned early, mice show high baseline levels of

glucocorticoids, as well as a stronger increase in response to stress (Kikusui, Nakamura, Kakuma, & Mori, 2006; Orbak, 2006). This results in abnormal regulation of the hypothalamus pituitary adrenal (HPA) axis, and increases anxiety like behaviours (Caldji et al., 1998). The myriad of abnormalities discussed so far highlight the global impact that early life stressors have on the brain and development. These changes are carried throughout the lifespan, affecting cognition, moods, social and stress responses (Caldji et al., 1998; Whishaw & Kolb, 2014).

Early life stress

Environmental stressors in early life show a relationship with psychopathological development. Children who are exposed to high levels of stress, such as deprivation or maltreatment are more likely to develop behavioural problems; which can progress into psychopathology in adulthood (Mogi et al., 2011). Early environmental stress can affect behavioural, emotional, cognitive and stress response development (Mogi et al., 2011). Parental care and maternal bond influence how a child copes with stress throughout their lifespan (Francis & Meaney, 1999). In the 1980's, a large scale adoption study took place of institutionalised Romanian children who had been severely deprived of early maternal care or social interaction (Rutter & O'Connor, 2004). Rutter and O'Connor (2004) found that early developmental deprivation correlated with cognitive and behavioural deficits; this was exacerbated by increased time in the deprived situation. The children showed a variety of cognitive and behavioural deficits, physical differences, and psychopathology, like anxiety and depression. The variation seen within these deficits suggests that the children's genetic makeup also plays a role during development, supporting theories on gene and environmental interactions, and the complex role that external and internal factors play on development (Caldji et al., 1998; Carola, Pascucci, Puglisi-Allegra, Cabib, & Gross, 2011).

Stress affects individuals in different ways, some will show resilience and others vulnerabilities in the face of stressful events (Canli et al., 2006). A genetic factor implicated in this variability is the length-polymorphism variation of the serotonin transporter gene (5-HTTLPR) which has been linked to genetic susceptibility and protection (Carola et al., 2011). These genetic and environmental factors interact to play a vital role in how we develop the skills to adapt or react to stressful situations. Around 15% of children suffer from serious anxiety problems, these children are at higher risk of developing psychopathologies in later life (Costello, Egger, & Angold, 2005). Roughly 29% of adults experience mental disorders sometime during their life (Steel et al., 2014), thus understanding the role that early life stress can have on development is important.

Animal models are an effective way to look at causal effects of maternal care on the development of such psychopathologies. In rodents, a mother's stress response has been proposed to be passed on to their pups through the mother's maternal care (Nephew & Murgatroyd, 2013). As rats environments are relatively stable across their lifespan, this may be a developmentally adaptive response, giving the offspring the tools to respond to their environment accordingly (Caldji et al., 1998). The maternal behaviour may act as environmental feedback which mediates the endocrine response to stress (Nephew & Murgatroyd, 2013). Research into the role of 5-HT in maternal care is sparse, it has been suggested that in mothers who displayed low licking and grooming maternal behaviours showed higher rates of 5-HT turnover than their high licking and grooming counterparts (Masís-Calvo, Sequeira-Cordero, Mora-Gallegos, & Fornaguera-Trías, 2013), suggesting a link between maternal care and 5-HT abnormalities. Other research has suggested that when injected with a 5-HT_{2C} receptor agonist (MK212), maternal behaviours are disrupted (Wu et al., 2016). This was predominantly found to interrupt the mother's motivation to collect her pups in the pup retrieval task. This provides further evidence for a relationship

between 5-HT and maternal behaviours. Antipsychotic medications, such as clozapine which are 5-HT₂ antagonists, have also been found to impact maternal care, interrupting pup approach, retrieval and nest building in mothers (Zhao & Li, 2009). These studies suggest a strong regulatory relationship between 5-HT and maternal behaviours.

Serotonin

The current study examines the relationship between the 5-HT systems and the level of maternal care. The serotonergic system is important for the regulation of mood, sleep, appetite and the development of emotional responses, such as anxiety (Hariri & Brown, 2006). A length-polymorphism variation of the 5-HTTLPR gene in humans has been suggested to influence depression (Carola et al., 2011). The two main variants implicated are the short versus long alleles, these are repeat polymorphisms that occur in the promotor region of the SLC6A41 gene, coding for the 5-HTT transporter. The variants have been found to moderate the effects of stressful experiences on the development of psychopathological disorders in different ways (Canli et al., 2006; Caspi et al., 2003). The short variant has been shown to increase vulnerability and the long to protect against the development of psychopathologies. This 5-HTT allele and environmental interactions have also been shown to affect the grey matter volume in the hippocampus and the amygdala. These areas are associated with the formation of social attachments and behaviours, implicating them in maternal attachment and care (Canli et al., 2006).

Genetic modifications in laboratory animals has led to animal models mimicking the genetic liability seen in humans with the short variant of the 5-HTT allele. The homozygous serotonin transporter (SLC6A41^{Hubr}) knockout rats (SERT^{-/-}) animals show high levels of anxiety and depression-like behaviours, whereas the heterozygous (SERT^{+/-}) show a milder phenotype similar to that seen in the human short/short 5-HTT allele (Holmes, Li, Murphy, Gold, & Crawley, 2003; Holmes, Yang, Murphy, & Crawley, 2002).

For the current study we will examine the two SERT variants above, along with a wild-type ($SERT^{+/+}$) as models of different 5-HTT allele functions. These $SERT^{-/-}$ animals used in the current study were bred using a reverse genetics approach, through ENU (N-ethyl-N-nitrosourea) direct target-selected mutagenesis in a wild-type Wistar rat (Smits et al., 2006).

Hypothesis

The current study examines the relationship between variations in the SERT gene ($SERT^{-/-}$, $SERT^{+/-}$, $SERT^{+/+}$) and the quality of maternal care provided by the mothers, the filial bond formed with their pups and their adult offspring's anxiety like behaviours.

Given the evidence implicating 5HTT in maternal care, we predict to see main effects of maternal genotype across multiple behavioural tests. We hypothesize that the $SERT^{-/-}$ will show poorer qualities of maternal care compared to both the $SERT^{+/-}$ and $SERT^{+/+}$. In addition, the offspring of the $SERT^{-/-}$ will show a weaker maternal bond and that the $SERT^{+/+}$ will show the strongest. We expect the anxiety-like behaviours of the adult offspring to reflect the quality of their mother's maternal care and bond, in that the $SERT^{-/-}$ will show higher levels of anxiety when compared to the $SERT^{+/-}$ and $SERT^{+/+}$.

We will also look at the relationship the maternal genotype has with the pup genotype; overall we expect the pup genotype to act as a protective or vulnerability factor in relation to the environmental stress of the mother's care. The $SERT^{-/-}$ offspring of $SERT^{-/-}$ mothers we predict to show the highest levels of anxiety as adults, in response to their mother's quality of maternal care and weak maternal bond. The $SERT^{+/+}$ offspring of $SERT^{+/+}$ and $SERT^{+/-}$ mothers we predict to show the lowest levels of anxiety as a response to high maternal care quality. We predict the $SERT^{+/-}$ offspring to sit somewhere in between the $SERT^{-/-}$ and $SERT^{+/+}$.

Method

Animals

Three female SERT genotype variants (6 SERT^{+/+}, 6 SERT^{+/-}, and 10 SERT^{-/-}) of the Wistar rat strain were used as breeding mothers. All were paired with SERT^{+/-} males to create a range of SERT genotype variants in the litters. Females and males were housed together for up to 21 days, or until the female was visibly pregnant. Males were removed from the cage and the females were housed individually to raise their offspring. The day of birth was counted as post-natal day zero (PND0), on which the mother was given shredded paper to encourage nesting and bonding with her pups. Litters were housed with mother until weaning on PND21. From this day pups were housed with sex matched litter mates, in groups of 2–6 animals depending on litter and animal size. Testing was carried out on PND4, 6, 11, 13, 40, 60.

The animals' day and night cycle was reversed so that the experiment took place during the rats' most active time (light on from 19:00 to 7:00). Food and water were available at all times, except when testing. All animals were housed at constant humidity (55%) and temperature (21±2°C), in conditions consistent with, and approved by the Animal Ethics Committee of Victoria University Wellington, and approved (AEC 22375).

Materials and Procedure

All animals were habituated to the room for 15 minutes prior to any testing. Four pups were chosen at random on the first day of testing on PND4, these animals were then tattooed and assigned a number from pup one-four based on the paw that was marked, these pups were used throughout testing. The remainder of the litter were not culled so as to minimise changes in maternal behaviours due to artificially small litter sizes.

Maternal Potentiation of Ultrasonic Vocalizations (USVs) of pups is a measure of the filial bond between the mother and pup. It is also an indirect measure of the level of

maternal care given to the pup by the mother, in that a stronger filial bond is a result of good maternal care (Hofer, Brunelli, Masmela, & Shair, 1996), (for further background information, please refer to appendix A).

After the initial habituation period the mother was separated from her litter and placed into a black (25x45x30cm) experimental box with no bedding. Mother was held in this box throughout the duration of testing. Pup one was then placed into a separate isolation chamber, a 10cm round container with a small amount of bedding at the bottom. This is set up approximately two meters from the mother's box (see figure 1).

Five minutes of USVs were recorded using Raven Pro Software (Cornell University, Ithaca, NY); this is isolation one (ISO1). In the reunion phase, the pup is reunited with the mother for 5 minutes, during which USV's and EthoVision (EthoVision XT 9; Noldus Information Technology, Leesburg, VA) video recordings were made. The mothers licking and rearing were both manually recorded as well. A second isolation period followed in which the pup was placed back in the isolation chamber for a further five minutes, USVs were recorded. This was then repeated for pup two. Pups three and four had the same isolation one and two phases, but were reunited with the litter as opposed to the mother for five minutes. Having one group reunited with the mother and the other with the litter permits the differentiation of general social interaction over specific maternal interaction. The overall number of calls a pup makes also reflects the maternal bond between the two. The maternal potentiation task was carried out on PND4 and 11 for all litters.

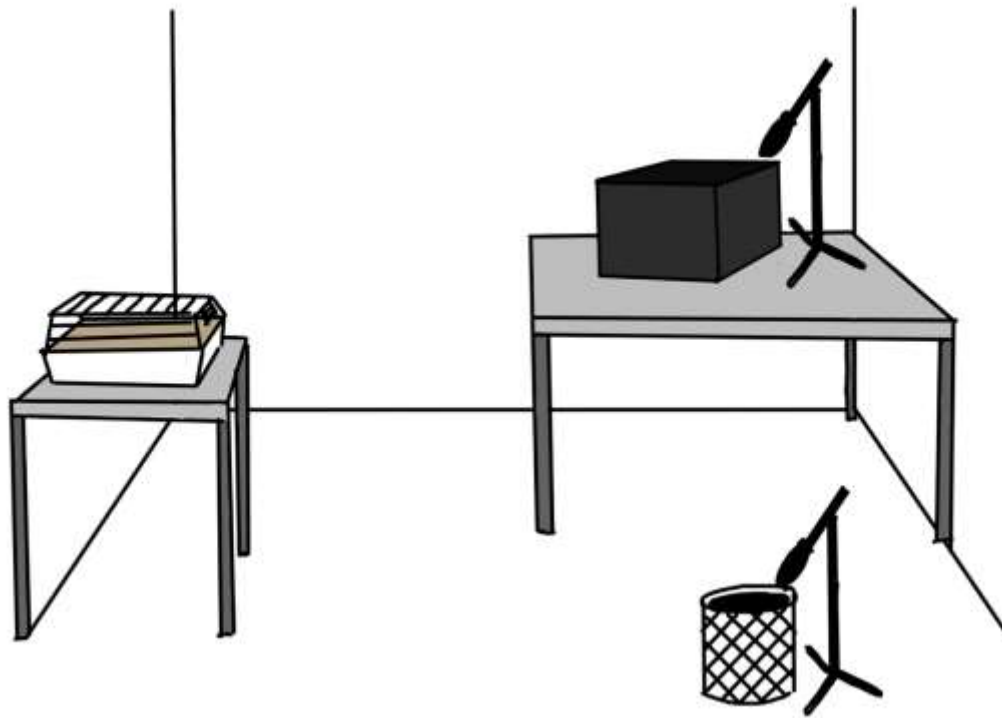


Figure 1. Room layout of the testing room for the maternal potentiation of USV's. From left; litter home cage, mother isolation box and microphone, pup isolation chamber and microphone in foreground. The pup isolation chamber was approximately 2 meters away from the mother's box.

The pup retrieval test was used as a measure of the mother's motivation to care for her pups (Capone, Bonsignore, & Cirulli, 2001). Following the initial habituation period, the mother was placed into a black experimental box (25x45x30cm) containing bedding from her home cage; for a 10-minute habituation phase. This was the same box which was used in the maternal potentiation experiment except with an attached second chamber (see figure 2). Four of the pups were placed in ramekins (two per ramekin) and placed in the separate compartment, with a perforated removable divider to isolate the mother from the pups. When the divider in place the mother wasn't able to access the pups, she was however able to hear and smell them. The task lasted for a total of six minutes and was video recorded using both EthoVision (EthoVision XT 9; Noldus Information Technology, Leesburg, VA).

After one minute had elapsed the divider was manually removed so that the mother had access to her pups, to retrieve at will. The latency for the mother to retrieve her pups was recorded. This was carried out on PND6 and 13 for all litters.

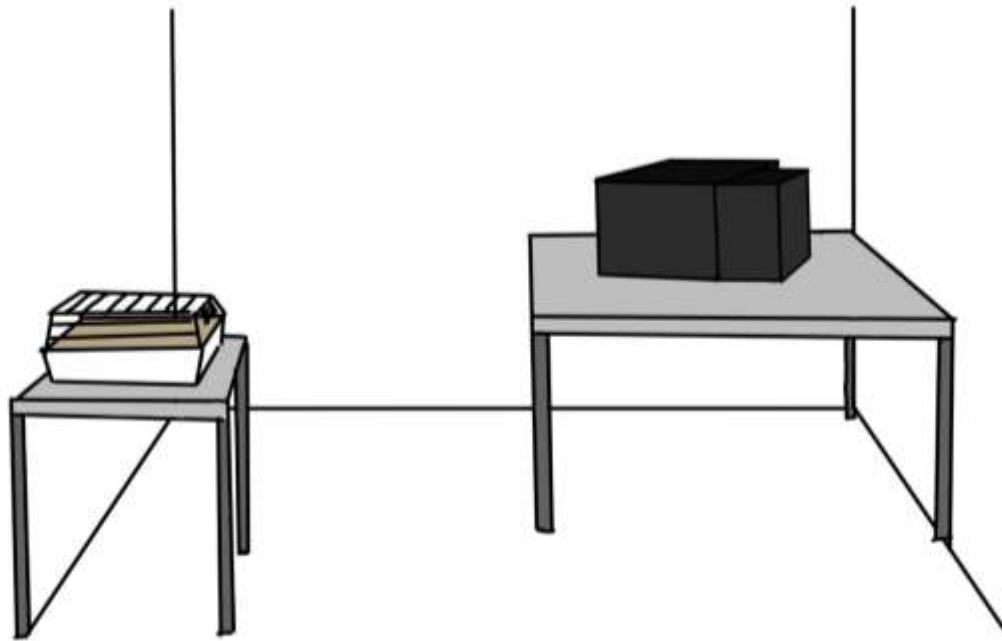


Figure 2. Layout of the testing room for the pup retrieval task. From left, the home cage of the remaining litter and the pup retrieval box in the background. Video recording was done through a webcam fixed to the ceiling.

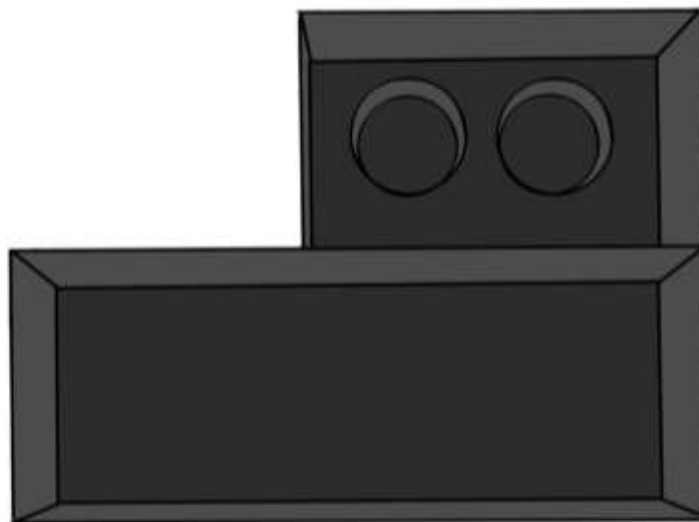


Figure 3. Chambers of the pup retrieval task. Larger chamber is where the mother was placed, and the partitioned off pup chamber includes two ramekins where the pups were placed. Partition between the two chambers was perforated and removable.

The SAT was used as a measure of anxiety behaviours (Deacon, 2013), (for further background information, please refer to appendix B). After the habituation phase, each animal was placed individually on the apparatus facing the back wall of alley one (see figure 3). Trials were recorded using EthoVision for a total of 5 minutes. Their latency to enter alleys two, three & four were recorded, as well as the cumulative time spent in each alley. Ceiling lights were on during the trials but were dimmed significantly. This was measured on PND40 for all litters.

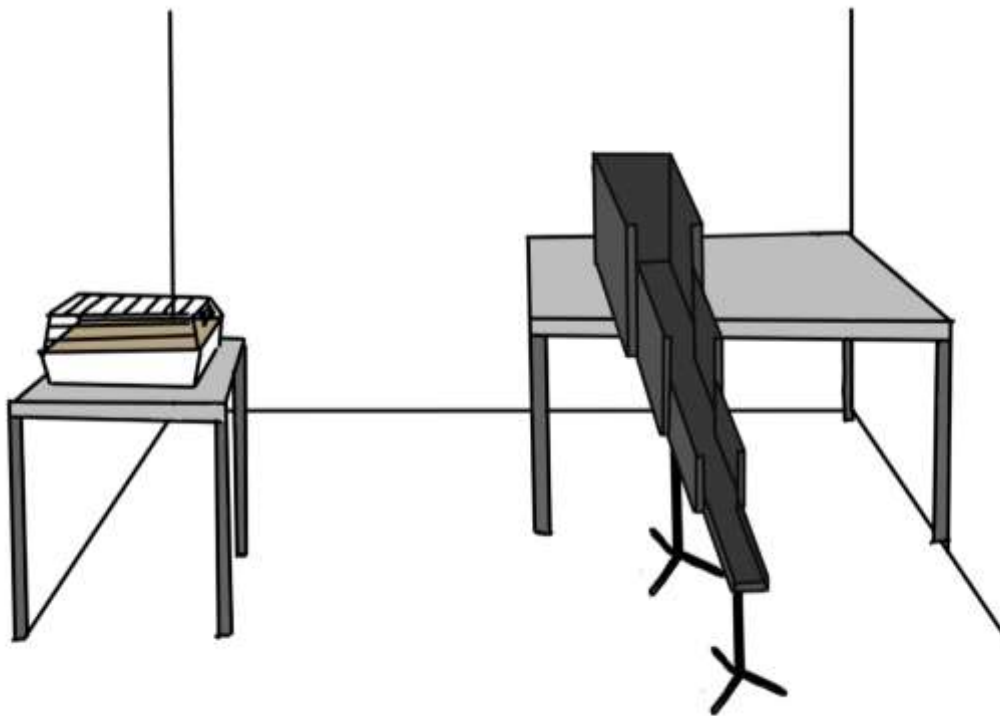


Figure 4. Layout of the testing room for the SAT. Including home cage (left) and SAT apparatus (right).



Figure 5. Birdseye and side on view of the SAT apparatus. Alleys progressively decrease in wall height and width. Alley 1, furthest left, to alley 4, furthest right.

Data analysis

All statistical analysis was performed using SPSS (IBM SPSS Statistics for Windows, Version 23.0. Armonk, NY: IBM Corp.). All tests were individually subjected to either a one-way or two-way analysis of variance (ANOVA). And were then subjected to post hoc Tukey tests. Significance was tested against a p-value of .05. Two pups were excluded from the maternal potentiation of USV's analysis as they died before being genotyped. These were both from the maternal $SERT^{-/-}$ genotype group (see table 1-3 for maternal descriptive statistics). Adult pups were added to the SAT analysis to give a broader spread across the pup genotypes, as pups were chosen prior to being genotyped, (for pup x maternal genotype descriptive statistics see appendix F and appendix G).

Table 1.

Descriptive statistics for maternal potentiation of USV's.

Maternal genotype	PND4			PND12		
	<i>M</i>	<i>SE</i>	<i>N</i>	<i>M</i>	<i>SE</i>	<i>N</i>
$SERT^{-/-}$	367.91	53.62	22	818.27	77.3	22
$SERT^{+/-}$	780.5	61.28	24	875.63	76.03	24
$SERT^{+/+}$	709.88	65.35	24	1072.38	79.59	24

Note: PND refers to postnatal day, *M* refers to mean, *SE* refers to standard error and *N* refers to sample size of group.

Table 2.

Descriptive statistics for pup retrieval test.

Maternal genotype	PND6			PND13		
	<i>M</i>	<i>SE</i>	<i>N</i>	<i>M</i>	<i>SE</i>	<i>N</i>
$SERT^{-/-}$	360	0	6	360	0	6
$SERT^{+/-}$	218.58	38.86	6	222.58	44.40	6
$SERT^{+/+}$	204.88	43.06	6	104.69	42.74	6

Note: PND refers to postnatal day, *M* refers to mean, *SE* refers to standard error and *N* refers to sample size of group.

Table 3.

Descriptive statistics for SAT.

Maternal genotype	PND6			PND13		
	<i>M</i>	<i>SE</i>	<i>N</i>	<i>M</i>	<i>SE</i>	<i>N</i>
SERT ^{-/-}	133.45	27.13	23	259.21	17.96	22
SERT ^{+/-}	117.20	21.41	27	163.18	25.15	27
SERT ^{+/+}	174.74	22.71	27	246.46	18.29	28

Note: PND refers to postnatal day, *M* refers to mean, *SE* refers to standard error and *N* refers to sample size of group.

Results

Maternal potentiation's of USVs, the difference between isolation one and two for the pups who were reunited with their mother was subjected to a one-way ANOVA, this showed no significant difference. As another approach we looked at the overall number of calls made across the two isolations, which is also used as a measure of the filial bond between the mother and pup (Hofer, Masmela, & Shair, 1999). The mean number of calls per maternal genotype were submitted to a two-way analysis of variance (ANOVA), for the PND4 and PND11 USV trials (for descriptive statistics see table 1). The analysis revealed that there was no significant interaction between maternal and pup genotypes in PND4 trial, $F(2, 63) = 0.04$, $p = .961$, or PND11 trial, $F(2, 63) = 1.14$, $p = .326$, (see appendix C for maternal and pup genotype figures).

PND4 Trial did not show a significant main effect of pup genotype $F(2, 63) = .48$, $p = .623$. But did show a significant main effect of maternal genotype, $F(2, 63) = 9.78$, $p < .001$, $\eta^2 = .237$. Further post hoc Tukey analysis showed that there was a significant difference between the SERT^{-/-} and the SERT^{+/+} ($p < .001$) and the SERT^{+/-} ($p < .001$); but no significant difference between the SERT^{+/+} and the SERT^{+/-} ($p = .495$). PND11 trial did not show a significant effect of maternal genotype, $F(2, 63) = 1.63$, $p = .205$, or pup genotype $F(2, 63) = .21$, $p = .810$. This result implies a trend in the same direction, whereby the SERT^{-/-} made less calls than both the SERT^{+/-} and SERT^{+/+} (see figure 6).

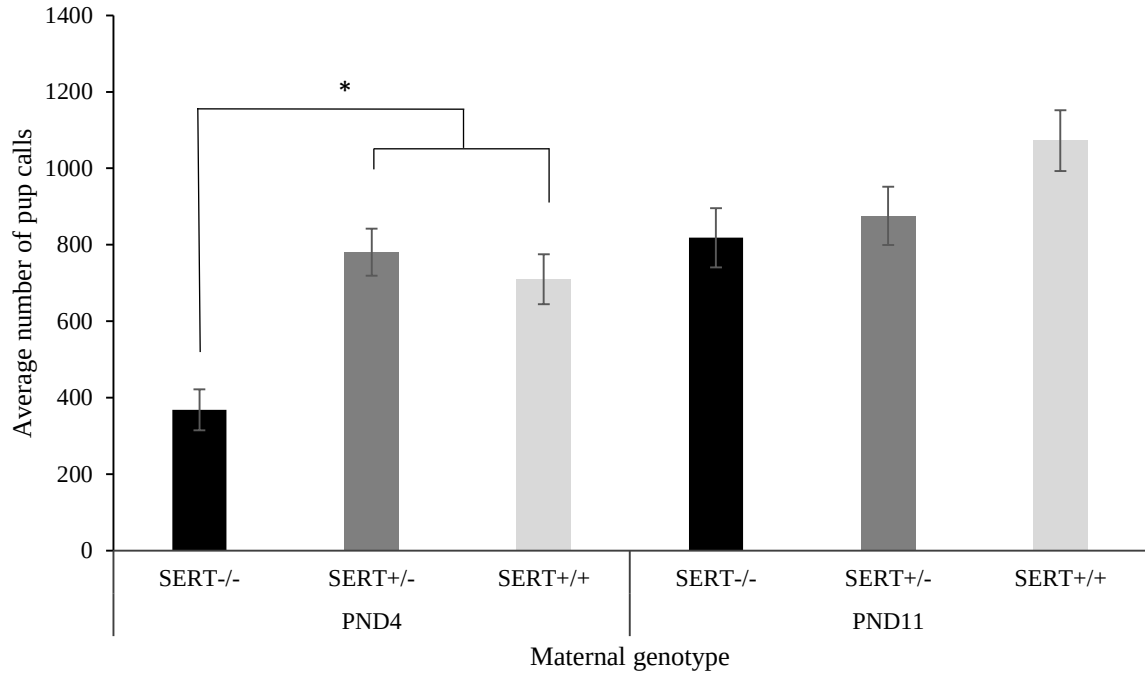


Figure 6. Differences in average number of calls made by pups during the maternal separation task during isolation one and two. Error bars depict standard error, PND refers to postnatal day.

Pup retrieval test, the average latency to retrieve all four pups from PND6 trial and PND13 trial of the pup retrieval test were independently subjected to a one-way ANOVA (for descriptive statistics see table 2). PND6 trial showed that there was a significant maternal genotype effect, $F(2, 15) = 6.58, p = .009, \eta^2 = .467$. Further post hoc analysis for PND6 trial showed that there was a significant difference between the SERT^{-/-} and the SERT^{+/+} ($p = .013$) and the SERT^{+/-} ($p = .024$), but no significant difference between the SERT^{+/+} and the SERT^{+/-} ($p = .955$). PND13 trial showed similar results, in that there was a significant effect of maternal genotype, $F(2, 15) = 6.87, p = .008, \eta^2 = .478$. Post hoc Tukey analysis showed that there was a significant difference between the SERT^{-/-} and the SERT^{+/+} ($p = .008$) and the SERT^{+/-} ($p = .039$), but no significant difference between the SERT^{+/+} and the SERT^{+/-} ($p = .705$). Suggesting that the SERT^{+/-} and SERT^{+/+} are similarly motivated to retrieve their pups, regardless of the different SERT genotype.

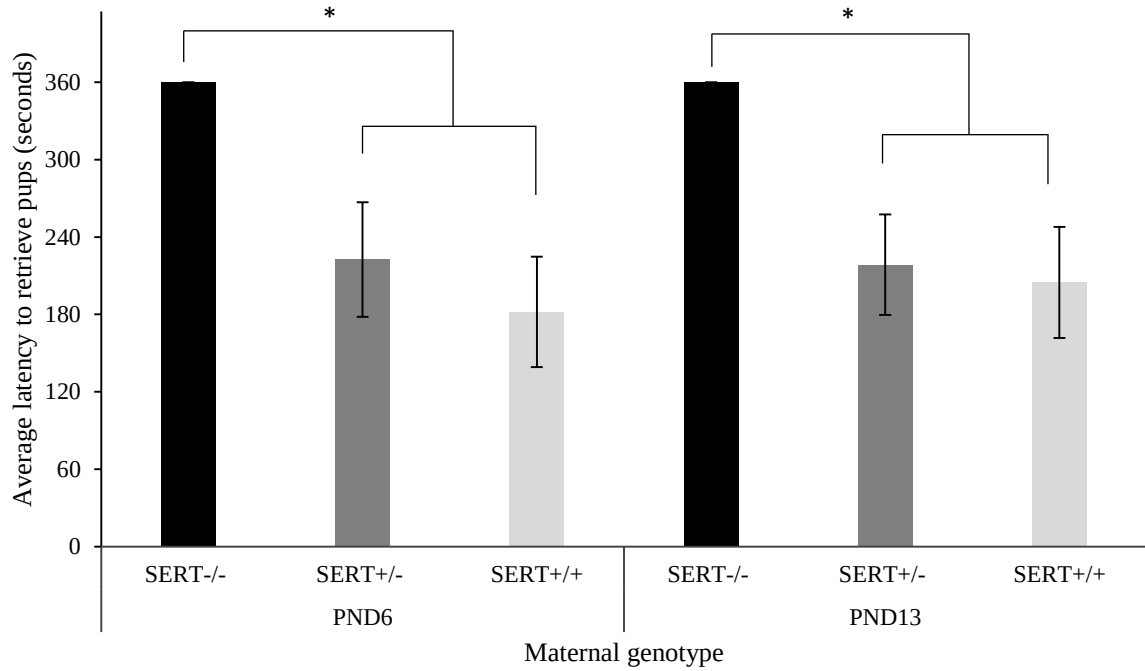


Figure 7. Average maternal latency to collect pups during the pup retrieval test. Error bars depict standard error. PND refers to postnatal day.

The SAT, mean latency to enter alley four was submitted to a two-way between subject's ANOVA. This was done for both PND40 trial and PND60 trial of the SAT, of maternal genotype (SERT^{-/-}, SERT^{+/-}, SERT^{+/+}) and pup genotype (SERT^{-/-}, SERT^{+/-}, SERT^{+/+}) (for descriptive statistics see table 3). The analysis revealed that there was no significant interaction between maternal and pup genotypes in PND40 trial, $F(2, 70) = .48$, $p = .623$, or PND60 trial, $F(2, 70) = .27$, $p = .766$, (see appendix D for maternal and pup genotype figures).

PND40 trial did not show a significant effect of maternal genotype, $F(2, 70) = 1.03$, $p = .361$, or pup genotype ($2, 70) = .55$, $p = .578$. PND60 trial did not show a significant effect of pup genotype $F(2, 70) = 2.12$, $p = .128$, but did show a significant effect of maternal genotype, $F(2, 70) = 5.48$, $p = .006$, $\eta^2 = .135$. Post hoc showed that there was a significant difference between the SERT^{+/-} and the SERT^{-/-} ($p = .007$) and the SERT^{+/+} ($p = .023$); but no significant difference between the SERT^{+/+} and the SERT^{-/-} (p

=.569). Suggesting higher levels of anxiety-like behaviours. Alternatively, it may be that the $SERT^{+/-}$ are more active than the $SERT^{+/+}$ and the $SERT^{-/-}$.

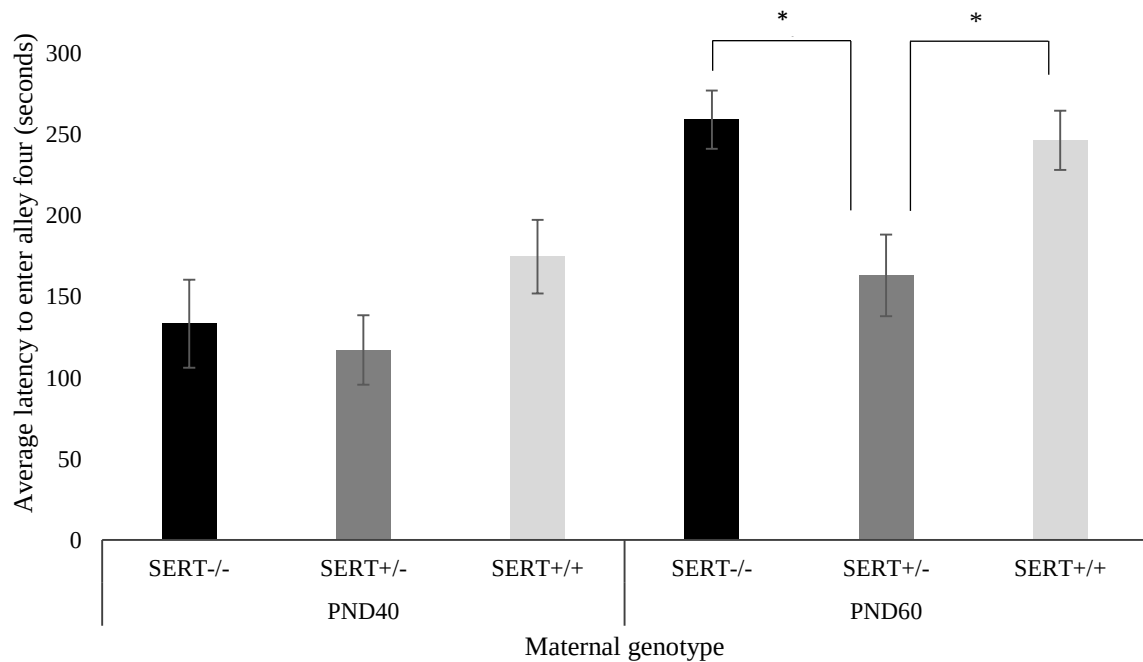


Figure 8. Average latency to enter alley four during the SAT, as organised by the maternal genotype. Error bars depict standard error. PND refers to postnatal day.

Discussion

The current study looked at the relationship between the quality of maternal care in relation to maternal SERT genotype, and the effects this has on the adult offspring's anxiety-like behaviours. We carried out three tests each at two time points, the potentiation of USV's and pup retrieval task were measures of maternal bond and care and the SAT was used as a measure of the grown pup's anxiety level.

Before going on to discuss the implications of the findings, it is important to highlight the difficulty we had with breeding the $SERT^{-/-}$ genotype. To have the six litters used for the current study, ten dams were bred. Of these, four of the females rejected their litter, resulting in the fatality of the litter. Of the six-surviving litters, several mothers consumed some of their pups (see appendix E). The results found in the current study therefore may overestimate the quality of maternal care provided by the $SERT^{-/-}$ mothers.

None of the SERT^{+/-} and SERT^{+/+} mothers rejected their litters and one SERT^{+/-} consumed one of her pups. This suggests that in our animals this is specific to the SERT^{-/-} genotypes.

It is commonly accepted that these behaviours are associated with maternal stress and survival. The cannibalism of pups by their mothers is a way of ensuring the remaining pups survival, through resource allocation (Taroco, Gaya, Resende, dos Santos, & Madureira, 2015). This is possible as mothers are able to instinctively assess the health of their pups, through olfactory and physical cues (Schardein, Petrere, Hentz, Camp, & Kurtz, 1978). This may result in different maternal care of the offspring, as such larger offspring will be given better maternal care as they are more likely to survive. Additionally, rejecting a litter may also be a reflection of the perception of stress in the environment, as SERT^{-/-} are highly sensitive to stress this may play a role in why they reject their litters (Taroco et al., 2015). If they perceive their environment as stressful, they may also see this as a reason why their litter is less likely to survive, and therefore not worth the energy to care for. By rejecting her litter, the SERT^{-/-} female can allocate the resources to herself and for future litters, and by consuming her pups she also gets the nutritional benefit. The SERT^{-/-} vulnerability to stress may play a role in their rejection and consumption of their litters. This also contributes to the evidence that there is a relationship between the SERT gene and maternal care.

Maternal potentiation of USV's. As a measure of the mother/pup bond, we used two isolation periods, however no difference between them was found. A possible explanation for this may lie in the pups inability to regulate their body temperature, resulting in a loss of body heat when separated from the mother and litter (Cohen, Rothschild, & Mizrahi). The loss of heat can affect the pup's calling, as their temperature decreases so does their calling (Blumberg & Alberts, 1991). As the second isolation occurs after a five-minute isolation, followed by a five-minute reunion, this may not be enough

time for the mother to bring the pups body back to temperature, approximately 35-36 °C, resulting in the pups making fewer calls. We combined the two isolations to look at the overall number of calls made by the pups as a measure of the filial bond. Future use of this paradigm may want to include a heating pad to mitigate the influence of body temperature.

The averaged total number of calls made by the pups showed that the $SERT^{-/-}$ on PND4 made substantially less overall calls than both the $SERT^{+/-}$ and $SERT^{+/+}$, suggesting they comparatively form a weaker mother/pup filial bond. This trend carries through to the PND11 trial, although it did not reach significance there is a similar overall pattern, with the $SERT^{-/-}$ making less calls. The failure to reach significance on PND11 may be that pups are more reliant on their mother's care at PND4 than PND11. Alternatively, may also be due to a lack of power, as the current sample size is not large enough to capture smaller effects. Overall we see a trend which shows a decrease in the number of calls made by the $SERT^{-/-}$ offspring.

Offspring USV's encourage mothers to care for their pups, suggesting that the pups themselves influence the maternal (Mogi et al., 2011). As we did not find a significant interaction between the maternal and pup genotype, we would speculate that the maternal care is largely driving this effect over the pup genotype. Perhaps, future research could investigate the mother's response to the pup calls, as well as cross-fostering the pups to look at changes in the offspring's USV's.

The pup retrieval test showed strong evidence that the $SERT^{-/-}$ were less motivated to retrieve their pups. In that none of the $SERT^{-/-}$ retrieved their pups, in either of the pup retrieval trials. The $SERT^{-/-}$ performance may be a reflection of their sensitivity to the stressfulness of the experimental paradigm. However, if we look at the $SERT^{-/-}$ data for the pup retrieval test, the number of USV's they made and the number of mothers who rejected their litters, we argue that this is compelling evidence that the $SERT^{-/-}$ are

susceptible to stress and this impacts the quality of their maternal care. Future research may want to look closer at the maternal behaviours of the SERT^{-/-} in relation to the nursing, licking, grooming and amount they sit on their nest, compared to the SERT^{+/-} and SERT^{+/+}.

The SERT^{+/-} and SERT^{+/+} were not significantly different from each other in the pup retrieval test. There was some variation in the average times of the SERT^{+/-} and SERT^{+/+}, this may be due to one outlier in each group, who didn't retrieve their pups. Overall the SERT^{+/-} and SERT^{+/+} were motivated to retrieve and care for their pups. This finding suggests that there is a link between the quality of maternal care and the SERT gene, and that this may be explained through a decreased motivation due to susceptibility to stress, or alternatively a decrease in the rewarding effects of social interactions (Kranz, Kasper, & Lanzenberger, 2010).

The SAT test on PND40 showed no significant differences between the three maternal genotypes, this may be due to increased locomotor activity and novelty seeking behaviours rats show at this time, and explain the overall fast latency to enter alley four for all genotypes (Hughes, 1968). This time translates onto the human timeline as adolescence, which is a developmentally sensitive period for the onset of anxiety disorders (Lijster et al., 2017). Like rats, adolescence in humans is a time when there is an increase in risk taking behaviours (Steinberg, 2004). Testing our animals during this period was based on the early onset of anxiety disorders, however the test may not be sensitive enough to test anxiety at this time due to the increase in risk taking behaviours.

The PND60 SAT trial showed that the SERT^{+/-} were significantly faster to enter alley four than both the SERT^{-/-} and SERT^{+/+}, who did not differ. A possible explanation for this finding is that the SAT is measuring different behaviours in the SERT^{-/-} and SERT^{+/+} genotypes. We speculate that the SERT^{-/-} offspring were highly anxious and their

slow latency reflects that. But that the SERT^{+/+} were less anxious, as the animals are able to see alley four they chose not to explore, as there was no novelty or reason. We did not measure the defecation of the animals but anecdotally found the SERT^{-/-} offspring defecated comparatively more, which is a sign of anxiety. We speculate that the test was measuring anxiety-like behaviours in the SERT^{-/-} and novelty seeking in the SERT^{+/+}. Preliminary studies from our laboratory using the SAT and food reward found that the SERT^{+/+} showed the fastest latency to enter alley four, and the SERT^{-/-} the slowest, this was the animals genotype as opposed to the maternal genotype used in the current study. Having the food reward as novelty may motivate the SERT^{+/+} to move out into alley four. This may be explored in future research, using novelty to test changes in their behaviours.

As the SAT is a conflict task, there is a balance between the desire to explore and the fear of open and high places, it may also be that the SERT^{+/-} are more explorative than the SERT^{-/-} and SERT^{+/+}, rather than an increase in anxiety in the SERT^{+/+}. Further exploration should be undertaken using a locomotive test to see if the SERT^{+/-} do show higher locomotive activity compared to the SERT^{+/-} and SERT^{+/+}.

General discussion

The pattern of our data suggests that there is a relationship between 5-HTT and maternal care. The mechanism behind this is not clear, as the strongest results were found in the mother's motivation to attend to her pups, we speculate the effect that 5-HTT has on maternal care may be a by-product of the relationship between 5-HTT and social reward (Kranz et al., 2010). Preliminary data from our lab has shown that the SERT^{-/-} show a decreased reward to social stimuli during adolescence, in that they do not perceive social interactions as rewarding. Using the social place preference paradigm (Calcagnetti & Schechter, 1992) to look at the rewarding effects of social interactions, where the animal is given the choice between a place where they have had a prior social experience, or one

that they had not. We found that the SERT^{-/-} are less likely to show a preference for the place where they had engaged in the social interaction. Data from a study looking at the sociability of SERT^{-/-} young shows that they engage less in play behaviours compared to the SERT^{+/+} (Homberg, Schiepers, Schoffelmeer, Cuppen, & Vanderschuren, 2007). These findings suggest both that the SERT^{-/-} are less likely to engage in social interactions, and when they do they find them less rewarding. These studies provide a possible explanation for the relationship between 5-HTT and maternal care found in the current study, based on the relationship that 5-HTT has with motivation and reward.

As the SERT^{-/-} are less motivated by social reward, this may weaken the effects of the maternal bond, through a decreased perception of the reward associated with caring for offspring. Pups have rewarding effects on the mothers which encourage maternal care, through olfactory, hormonal and physiological interactions of both the mother and the offspring (Ferris et al., 2005). This is important for the survival of the litter and the motivation of the mother to look after them. As the SERT^{-/-} are less responsive to social rewards, they may not feel the rewarding effects of these systems, resulting in them showing less motivation to care for their offspring. This is speculative and requires further exploration to understand the mechanisms behind why we see a lower quality of maternal care in the SERT^{-/-} mothers. A deficit in the rewarding feeling of social interactions and how that impacts the mother's motivation to show maternal care would provide a research avenue to begin to understand this relationship. There is evidence for this in the moderating role that 5-HTT has with the dopaminergic and norepinephrine systems, these are both associated with the reward systems in the brain (Bearer, Zhang, Janvelyan, Boulat, & Jacobs, 2009). In a study looking at the brain circuitry functioning of SERT^{-/-} mice, using magnetic resonance methods Bearer et al. (2009) showed that the pathway from the prefrontal cortex to the limbic system was disrupted. These systems are

associated with reward and motivation (Bearer et al., 2009). Alteration in 5-HT levels influences reward processing in a number of ways, such as sensitivity, discrimination, and prediction (Kranz et al., 2010). This further implicates the modulating role of 5-HT in the reward systems, and may play a role in why we see the SERT^{-/-} mothers providing lower quality of maternal care in the current study.

As there were no significant differences in the SERT^{+/-} and SERT^{+/+} genotypes in their quality of maternal care as seen by the pup retrieval test and the pup's USVs, this suggests that there is an optimum level of 5-HTT, whereby an increase does not result in better care.

Conclusion

The current study shows that there is a link between the quality of maternal care and the SERT gene in rats. This may be due to the moderating effects that 5-HTT has on the reward and motivational systems in the brain and the decreased social reward that the SERT^{-/-} feel. We also suggest that there is an optimal level of 5-HTT whereby further increases have no effects on the quality of maternal care as seen in the similarity between the SERT^{+/-} and SERT^{+/+} genotypes. This is important research as it begins to tease apart the effect that SERT has on maternal care. It also provides evidence that the maternal background of lab animals is important to take into account when using them for research purposes.

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Appendix A

Ultrasonic vocalisations were used as sound is an important sense during development, it plays a key role in communication and maternal bonding (Nakayama, 2015). When a human infant cries their mother will respond, by providing maternal care, strengthening attachment. Cries in this way are often biologically driven in early infancy but as they develop this shifts to be socially driven (Nakayama, 2015). This is the same for a number of mammals including rats. USVs illicit maternal behaviours by the mother, of licking, nesting and retrieval behaviours, in this way similar to human infant cries, pup calling encourages interactions between the pup and the mother and typically convey negative effective states in the pup (Mogi et al., 2011; Shair et al., 2015). This makes USV a good instrument for indirectly measuring maternal care as well as filial bonding, based on the paradigm used. As an extension of this, the mother's maternal instinct, motivation, and quality of care is seen in the prompt retrieval of her pups when they stray from the nest (Capone et al., 2001), this is often initiated through the pups distress calling and is driven by the mothers desire to keep her pups safe. These early environmental factors and stresses contribute to the development of the pup, those whose mothers provided good quality maternal care have been shown to display less anxiety-like behaviours (Stolzenberg, Stevens, & Rissman, 2012).

Appendix B

The SAT used in the current experiment as a measure of anxiety and the procedure followed was outlined by Deacon (2013) with some alterations. The apparatus consists of 4 alleys decreasing in wall height (alley 1 = 25cm, alley 3 = 10cm, alley 2 = 3cm, alley 1 = 1cm), and in width (alley 1 = 8.5cm, alley 3 = 8.5cm, alley 2 = 3.5cm, alley 1 = 1.2cm). This gives the animals a gradual exposure to the higher anxiety evoking alleys. As opposed to a gradual colour change the apparatus used was black for ease of tracking and video recording. The SAT is argued as a more sensitive measure of an animal's anxiety level, as the apparatus exposes the animals to increasing levels of anxiety evoking heights and open space those who venture out into the furthest alleys show increased risk behaviours and therefore lower level of anxiety. Animals who spend more time in alley 1 exhibit high levels of anxiety, those who venture into alley 2 show moderate levels of anxiety and those who venture out into alleys 3 and 4 show low levels of anxiety.

Appendix C

The average number of USV's made by pups during the potentiation of USV's trials, as organised by maternal and pup genotype.

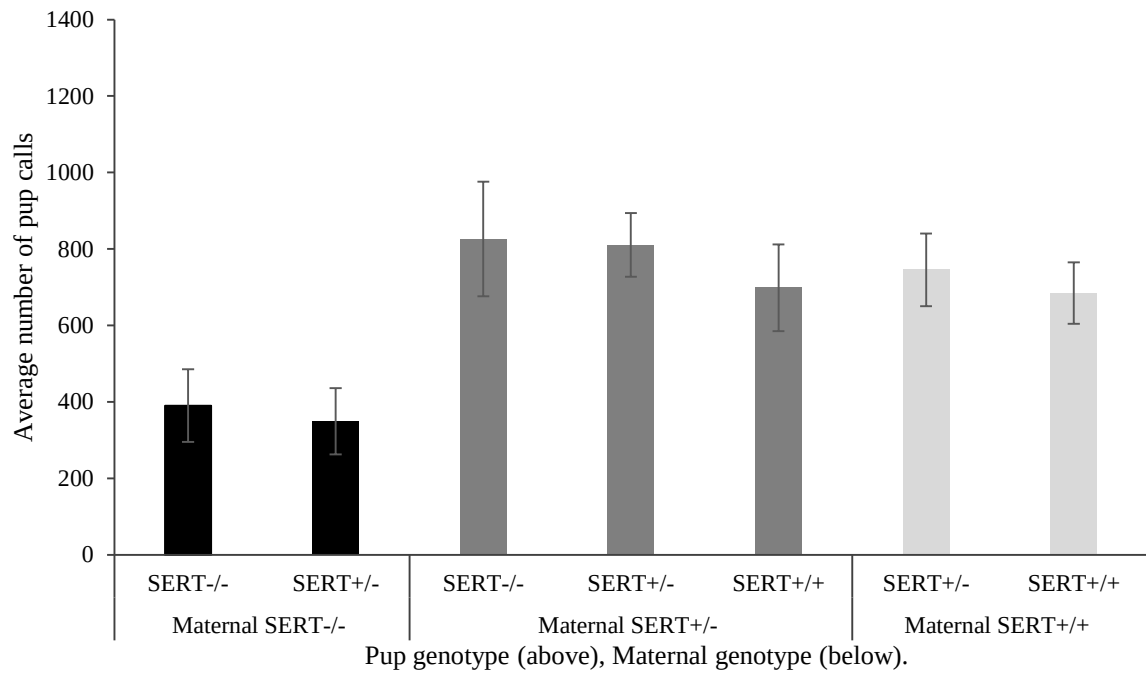


Figure 9. Average number of USV's made by pups during the PND4 maternal separation task. Values depict the combined average of pup isolation one and two. As sorted by maternal and pup genotype. Error bars depict standard error.

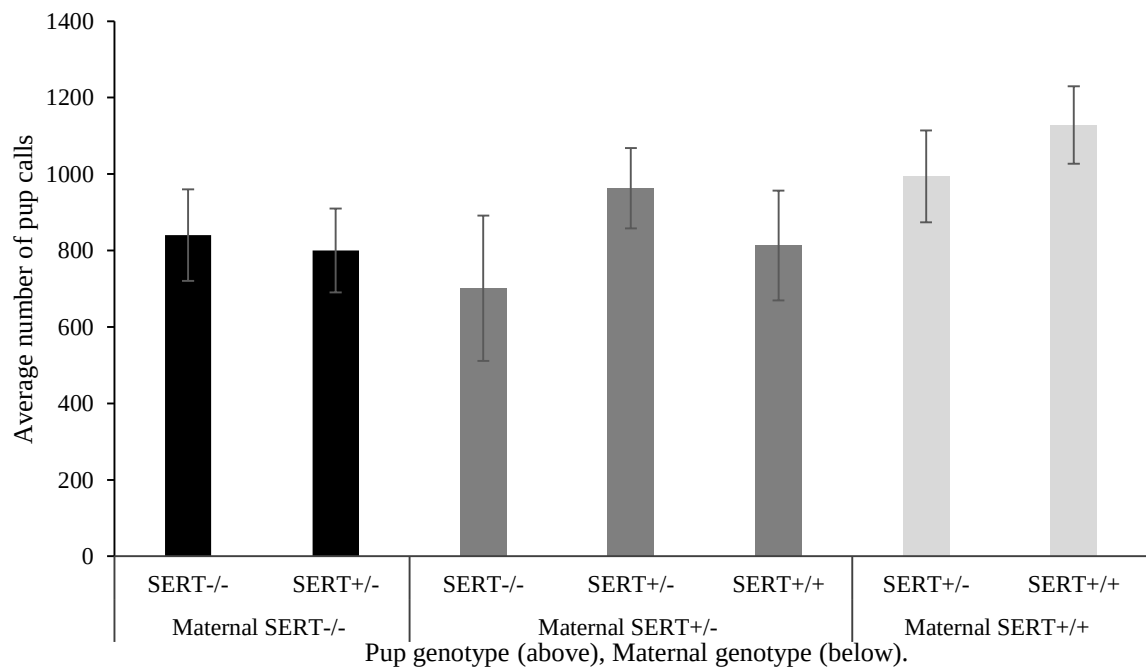


Figure 10. Average number of USV's made by pups during the PND11 maternal separation task. Values depict the combined average of pup isolation one and two. As sorted by maternal and pup genotype. Error bars depict standard error.

Appendix D

The average latency to enter alley four during the SAT, as organised by maternal and pup genotype.

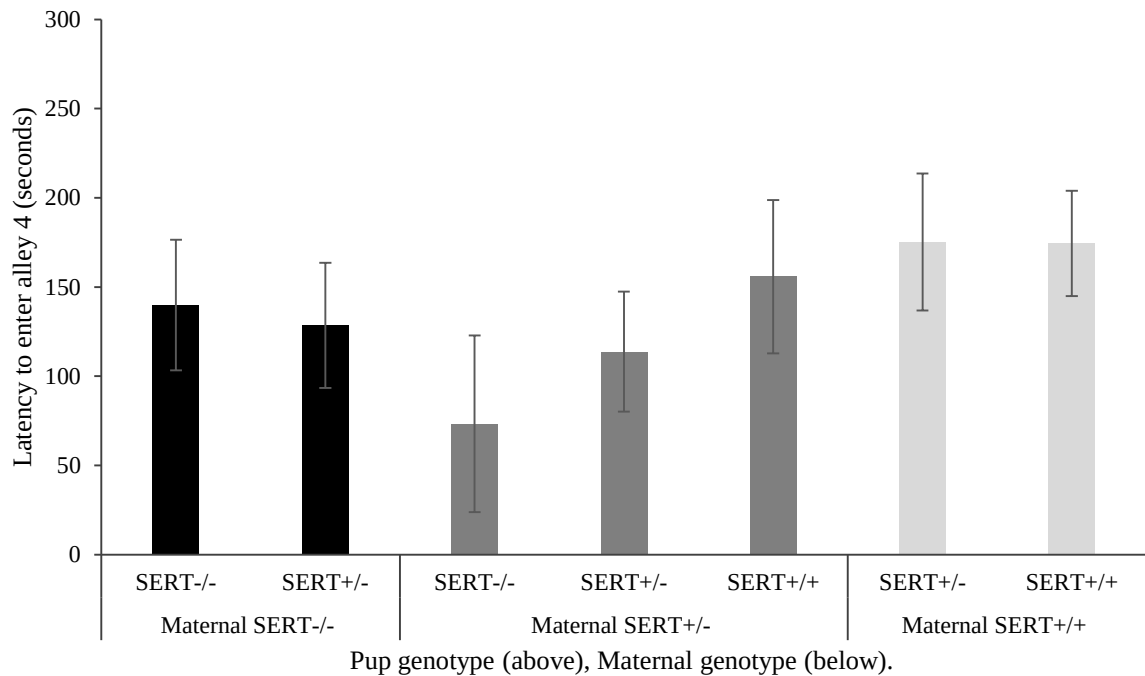


Figure 11. Average latency to enter alley four during PND40 SAT. Organised by the maternal and pup genotype. Error bars depict standard error.

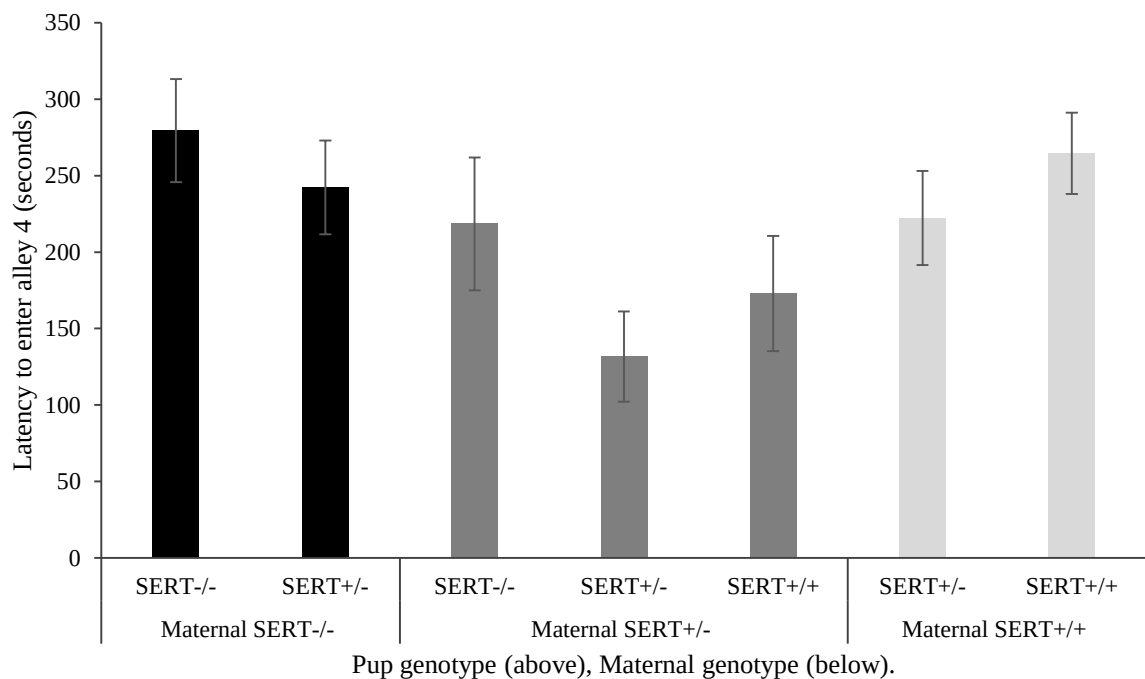


Figure 12. Average latency to enter alley four during PND60 SAT. Organised by the maternal and pup genotype. Error bars depict standard error.

Appendix E

Table E 1.

Total number of pups per litter and those that died.

Litter no.	SERT-/-			SERT+/-			SERT+/+		
	No. of pups	Died PND4	Died PND 19-24	No. of pups	Died PND4	Died PND 19-24	No. of pups	Died PND4	Died PND 19-24
1	9	5		8			9		
2	7	2		12			9		
3	—	—	—	12	1		12		
4	9		7	11			12		
5	—	—	—	7			11		
6	13	1	7	10			14		
7	—	—	—						
8	9	2							
9	11	7							
10	8								

Note: No. refers to number. — refers to litters not counted, as the entire litter had been rejected. All animals that died before PND4 were either rejected or consumed by the mother. Those that died between PND19-24 died of unknown causes.

Appendix F

Table F 1.

Descriptive statistics for pup and maternal genotype, maternal potentiation of USV's PND4.

Maternal genotype	Pup genotype	<i>M</i>	<i>SE</i>	<i>N</i>
SERT ^{-/-}	SERT ^{-/-}	390.1	94.79	10
	SERT ^{+/-}	349.42	86.53	12
SERT ^{+/-}	SERT ^{-/-}	826.25	149.87	4
	SERT ^{+/-}	810.54	83.14	13
	SERT ^{+/+}	698.57	113.29	7
SERT ^{+/+}	SERT ^{+/-}	745.4	94.79	10
	SERT ^{+/+}	684.5	80.11	14

Note: M refers to mean number of calls made, *SE* refers to standard error, *N* refers to number in group.

Table F 2.

Descriptive statistics for pup and maternal genotype, maternal potentiation of USV's PND11.

Maternal genotype	Pup genotype	<i>M</i>	<i>SE</i>	<i>N</i>
SERT ^{-/-}	SERT ^{-/-}	840.2	120.09	10
	SERT ^{+/-}	800	109.63	12
SERT ^{+/-}	SERT ^{-/-}	701.25	189.88	4
	SERT ^{+/-}	963	105.33	13
	SERT ^{+/+}	813	143.54	7
SERT ^{+/+}	SERT ^{+/-}	994	120.09	10
	SERT ^{+/+}	1128.36	101.49	14

Note: M refers to mean number of calls made, *SE* refers to standard error, *N* refers to number in group.

Appendix G

Table G 1.

Descriptive statistics for pup and maternal genotype, SAT PND40.

Maternal genotype	Pup genotype	<i>M</i>	<i>SE</i>	<i>N</i>
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SERT ^{-/-}	SERT ^{-/-}	138.86	36.6	11
	SERT ^{+/-}	128.5	35.05	12
SERT ^{+/-}	SERT ^{-/-}	73.3	49.57	6
	SERT ^{+/-}	113.72	33.68	13
SERT ^{+/+}	SERT ^{+/+}	155.77	42.93	8
	SERT ^{+/-}	175.25	38.4	10
	SERT ^{+/+}	174.43	29.45	17

Note: M refers to mean number of calls made, *SE* refers to standard error, *N* refers to number in group.

Table G 2.

Descriptive statistics for pup and maternal genotype, SAT PND60.

Maternal genotype	Pup genotype	<i>M</i>	<i>SE</i>	<i>N</i>
SERT ^{-/-}	SERT ^{-/-}	279.51	33.68	10
	SERT ^{+/-}	242.3	30.74	12
	SERT ^{-/-}	218.48	43.48	6
SERT ^{+/-}	SERT ^{+/-}	131.71	29.54	13
	SERT ^{+/+}	172.85	37.65	8
SERT ^{+/+}	SERT ^{+/-}	222.28	30.74	12
	SERT ^{+/+}	264.6	26.62	16

Note: M refers to mean number of calls made, *SE* refers to standard error, *N* refers to number in group.